



Perspective

Hallucinations in generative AI: A threat to scholarly integrity and the urgent need for publisher-led academically supervised verification

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ARTICLE INFO

Keywords:

AI hallucinations
Fabricated citations
Probabilistic token prediction
Scholarly integrity
Restricted jurisdictions
Reference checking

ABSTRACT

Generative AI systems, such as chatbots and AI-enhanced search engines, commonly produce hallucinations—generating smooth, convincing text that is factually wrong or entirely made up—particularly when responding to specific, specialized academic questions. This poses a direct threat to fields that depend on precise, trustworthy literature, including energy research, where inaccuracies in battery degradation rates, solar efficiency figures, or carbon capture costs can have significant real-world consequences for infrastructure investment and climate policy. This letter examines AI hallucinations while intentionally avoiding any claims of intentional or targeted errors. The core argument is twofold: first, hallucinations require urgent publisher-led verification systems; second, these same technologies risk becoming instruments of control unless governed transparently. These hallucinations arise from the models' core mechanism of probabilistic next-token prediction, rather than genuine information retrieval, which causes them to invent details like nonexistent papers, fabricated DOIs, journals, and supporting narratives whenever training data is limited, outdated, contradictory, or insufficient. The problem remains widespread across different AI providers and continues to affect even systems augmented with retrieval mechanisms, often resulting in substantial rates of fabricated citations. Although this is not a form of deliberate misinformation, it generates chains of errors that undermine the reliability of scholarly work. To address the issue effectively in the long term, academic publishers should implement mandatory AI-assisted reference validation during manuscript submission, integrating real-time database checks, grounding methods, and human editorial oversight. Such publisher-driven verification is critical to reducing the spread of AI-generated inaccuracies and preserving confidence in the integrity of scientific literature. However, the threat of AI censorship of undesired content requires the scientific community to be extremely vigilant to retain decisive stewardship of the scholarly record.

1. Introduction to hallucinations in generative AI

This paper advances a dual thesis: (1) AI hallucinations pose a genuine threat to scholarly integrity that requires proactive publisher-led verification systems; (2) the same technologies, if governed without transparency, risk becoming instruments of control rather than integrity safeguards. Each section below links back to this core argument. Generative AI tools, including chatbots and AI-powered search engines, frequently hallucinate—producing fluent but factually incorrect or fabricated content—especially on precise, niche academic queries. This unreliability stems from hallucination, a well-documented emergent behavior in large language models (LLMs) where outputs are fluent and syntactically coherent yet factually inaccurate, unsupported, or entirely fabricated (Alansari and Luqman [1]; Huang et al. [2]; Samanta & Chakraborty [3]). The energy sector exemplifies the stakes: a

hallucinated citation about a novel battery chemistry's cost or a carbon capture system's efficiency could skew meta-analyses, misdirect research funding, or inform flawed policy decisions. To understand why this happens, it is essential to grasp the fundamental mechanism of these models. LLMs are not databases that retrieve information; they are complex statistical engines trained on vast amounts of text. Their core function is to predict the next most probable word (or “token”) in a sequence, based on the patterns they have learned. When prompted with a question, they generate a response by autoregressively building it one token at a time, always choosing the statistically most plausible continuation. This process optimizes for fluency and coherence. However, it has no inherent mechanism for verifying truth against an external source. When the training data lacks sufficient information on a niche topic, or contains conflicting or outdated information, the model's statistical patterns will inevitably fill the gap with the most plausible-

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<https://doi.org/10.1016/j.erss.2026.104720>

Received 18 February 2026; Received in revised form 14 April 2026; Accepted 15 April 2026

Available online 17 April 2026

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sounding, but potentially fabricated, details to complete the response. This is the architectural root of hallucination. This issue has been extensively studied since early models like GPT-3 and persists across providers (OpenAI, Google, Anthropic, Deep Seek, and others), even in retrieval-augmented generation (RAG) systems intended to ground outputs in external sources (Kalai et al. [4]; Magesh et al. [5]).

Fig. 1 presents the causal and remedial architecture of AI hallucinations in scholarly work as a horizontal three-stage flow:

- WRITING stage (blue, §§2–4): Probabilistic token prediction (§2) + data limitations (§4) → hallucinations (§§2–3) → compound deceptions (§3) and geographic inconsistencies (§4).
- REVIEWING stage (orange, §§6–7): Error cascades via paper mills, predatory journals, and citation networks (§7) → erosion of trust across energy, medicine, law, and AI/CS (§6).
- PROCESSING stage (green, §§10–15): Publisher-led verification (§§10,12) + hybrid human–AI review with transparent oversight and appeals (§§13–14) → integrity, equitable access, academic freedom (§15).

Arrows indicate causal flow left to right. Section numbers in each node guide readers to the detailed discussion.

2. Evidence of hallucinated references and content in academic use

As illustrated in the WRITING stage of Fig. 1, the probabilistic mechanisms of LLMs directly produce fabricated outputs. Generative AI models, such as ChatGPT, present substantial risks to academic and scientific research by generating hallucinations—fabricated or inaccurate references and content that appear plausible but lack factual basis. In one study, ChatGPT produced 178 references for research proposals, of which 69 lacked valid DOIs (with 41 traceable via internet search, indicating partial hallucination) and 28 were neither identifiable by DOI nor locatable through Google search (Athaluri et al. [6]). Another analysis of 115 references generated by ChatGPT-3.5 in medical content revealed that 47% were entirely fabricated, 46% cited real sources but contained inaccurate details, and only 7% were both authentic and accurate; the model frequently reinforced errors when challenged (“double-downing”) (Bhattacharyya et al. [7]). A biomedical case illustrated this issue when ChatGPT fabricated a nonexistent paper on tick egg-laying behavior, including invented authors, DOI, and journal details—confirmed as false by contacted researchers—prompting strong cautions against relying on such outputs without thorough verification (Goddard [8]).

3. High-profile examples in premier AI conferences

This section continues the WRITING stage analysis, showing that even rigorous peer review (the REVIEWING stage) fails to catch hallucinations without dedicated tools. A striking recent example underscores the scale of the problem. In January 2026, GPTZero’s analysis of 4841 papers accepted to NeurIPS 2025—the premier AI conference—revealed at least 100 confirmed hallucinated citations spanning 53 papers (approximately 1% of accepted work), despite rigorous peer review by 3–5 experts per submission (Shmatko et al. [9]). These included fully fabricated papers with invented authors, titles, journals, and DOIs, as well as subtle blends of real and fake elements. A follow-up taxonomy of these “compound deceptions” showed how hallucinations evaded detection and entered the elite scholarly record (Ansari [10]). Similar incidents occurred at ICLR 2026 under-review papers, confirming the issue is systemic and accelerating in 2025–2026 (Shmatko et al. [9]). Such cases demonstrate that even domain-expert reviewers cannot reliably catch AI-generated fabrications without dedicated tools.

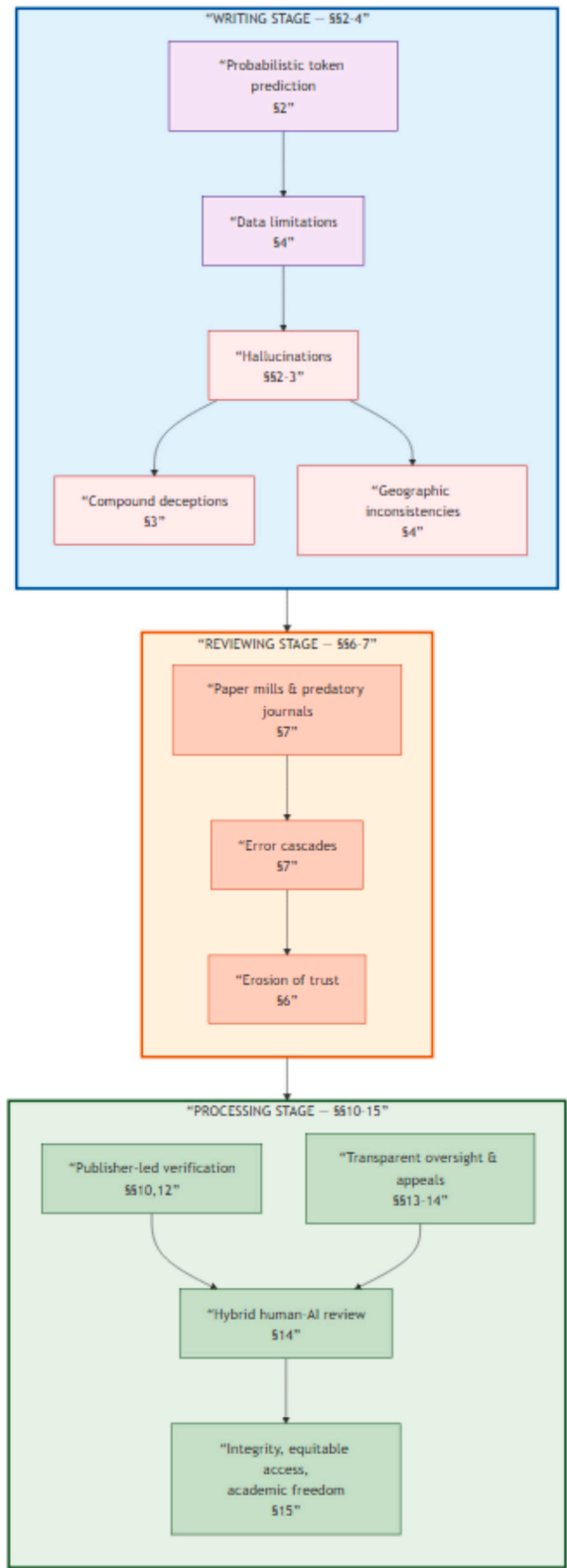


Fig. 1. Full causal and remedial architecture of AI hallucinations in scholarly work: from probabilistic token prediction through error cascades to publisher-led verification and integrity preservation. Stages are color-coded (blue: WRITING, §§2–4; orange: REVIEWING, §§6–7; green: PROCESSING, §§10–15). Arrows indicate left-to-right causal flow. Section numbers in each node direct readers to the corresponding text. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4. Geographic inconsistencies and their role in exacerbating hallucinations

In a striking preliminary investigation I performed recently, the query “*find a paper by [specific author – myself] on [specific subject – ‘green’ colonialism]*” produced dramatically divergent results from the same underlying model family, differing only by deployment region. In one jurisdiction, the system invented a completely nonexistent paper—including fabricated authors, title, journal, year, volume, pages, DOI, and a detailed narrative backstory. The described paper does not exist in any academic database or index, and the accompanying narrative was entirely invented. In another jurisdiction, it correctly identified the genuine work, contextualized its background, and suggested related papers (though with a minor error in volume numbering, an “*approximation*” artifact commonly seen in AI-generated citations where non-critical metadata is probabilistically generated rather than retrieved; Walters & Wilder, 2023; Magesh et al., 2025). Such geographic inconsistencies arise from variations in data-center indexing, knowledge cutoffs, regional content filters, A/B testing, and compliance with local data-localization or censorship requirements. These geographic divergences are not merely technical artefacts; they highlight how data restrictions—whether from censorship, localization laws, or incomplete indexing—exacerbate hallucinations. In environments with limited global scholarly access, models default more aggressively to pattern completion, fabricating content to fill voids.

5. Core architectural mechanisms and tradeoffs

The underlying mechanism causing AI system failures is straightforward: when prompted to “*find*,” “*cite*,” or “*rewrite with sources*,” LLMs pattern-match against learned bibliographic styles. For niche, recent, or low-frequency items, they probabilistically synthesize realistic-looking elements (plausible journal names, DOIs, abstracts) rather than admitting uncertainty. Rewriting tasks amplify this drift; even RAG systems can distort partial retrievals through the generative layer. Outcomes range from wholly fabricated citations to subtle inaccuracies (wrong volume, misattributed claims) or invented contextual elaboration (Walters and Wilder [1]; Samanta & Chakraborty [3]). This is usually a core architectural tradeoff: LLMs are optimized for fluency and coherence over strict factual fidelity. Leading labs actively mitigate the problem through RAG grounding, reinforcement learning from human feedback (RLHF) to penalize fabrications, uncertainty estimation, citation-verification sub-models, and continuous retraining. These efforts have produced measurable improvements, yet hallucinations remain stubbornly persistent on edge cases such as recent publications, controversial topics, or sparse scholarly domains. Hallucinations arise from core architectural tradeoffs in LLMs, optimized for fluency over strict factual fidelity rather than deliberate deception, and can be mitigated through probabilistic uncertainty estimation such as semantic entropy (Farquhar et al. [11]). Semantic entropy measures the model's uncertainty about the meaning of its own outputs, flagging potentially fabricated passages. It can be applied during inference (real-time, while the model generates) or post-hoc (on already-generated text), though post-hoc detection is less reliable because the model's internal uncertainty states are no longer accessible. For verification in scholarly publishing, real-time application during manuscript preparation or submission-stage screening is preferred.

6. Persistent hallucination rates in recent models and high-stakes domains

Returning to the REVIEWING stage of Fig. 1, error cascades from uncorrected hallucinations erode trust across disciplines. Recent 2025–2026 studies confirm that even advanced models like GPT-4o and Claude 3.7 still exhibit 15–20% hallucination rates on factual citation tasks, rising sharply (to 35–55%) on niche or recent topics (Magesh et al.

[5]; Vectara [12] [13]). In medical and legal domains—fields demanding absolute precision—rates can exceed 28% without grounding, with real-world consequences including sanctioned court filings and retracted preprints (Samanta & Chakraborty [3]). The energy sector is another such high-stakes domain. Research in this field directly informs multi-billion-dollar infrastructure investments, climate policy, and assessments of grid reliability and energy security. A hallucinated citation about the degradation rate of a new solar photovoltaic material, the projected cost of a novel battery chemistry, or the historical accuracy of a climate model could lead to poor investment decisions, flawed policy, or a misdirection of future research efforts. For instance, an AI tool tasked with summarizing the state-of-the-art in carbon capture could easily fabricate a non-existent study reporting an implausibly high efficiency rate (e.g., 95% capture at \$10/ton), which, if uncritically accepted, could skew an entire meta-analysis or mislead policymakers. The reliance on AI for literature reviews in energy transitions research thus carries a significant and under-appreciated risk. The systemic consequences are significant. While LLMs excel at synthesis, ideation, and common-knowledge tasks, they routinely fail in precision-critical areas like academic referencing or recent scientific developments. AI-augmented search often wraps web snippets in fluent prose that can still invent or distort facts. Fabricated citations spread rapidly through manuscripts, preprints, and media, creating error cascades that undermine scholarly integrity. Recent analyses confirm that GPT-3.5 fabricated 55% of citations in generated literature reviews, while GPT-4 reduced this to 18%—a clear advance, yet still unacceptable for rigorous work. Even specialized RAG tools (e.g., legal-research systems) hallucinate 17–34% of the time on benchmark queries, far below general-purpose claims of “*hallucination-free*” performance (Walters and Wilder [14]; Magesh et al. [5]).

7. Broader manifestations and propagation in the scholarly ecosystem

Beyond citations, hallucinations now manifest as fabricated data summaries, invented experimental outcomes, and entire “*paper-mill*” style manuscripts. In 2025–2026, elite venues like NeurIPS saw verified breaches of hallucinated citations, while the broader issue of AI-generated content in lower-tier and predatory journals has also been widely reported (Shmatko et al. [9]). These errors propagate through citation networks, inflating h-indices with ghost references and skewing meta-analyses. The proliferation of AI-generated content has particularly fueled the growth of paper mills—fraudulent operations that mass-produce low-quality, plagiarized, or entirely fabricated manuscripts for sale to authors seeking easy publications—and amplified their infiltration into predatory journals. In 2025–2026, reports indicate that predatory publishers, already numbering over 20,000 globally, have seen increased submissions of AI-assisted or fully synthetic papers, which exploit minimal or sham peer-review processes to generate revenue while further polluting the scholarly record (Cabells [15]). These outlets often host redundant “*copycat*” papers that paraphrase existing studies with AI tools to evade traditional plagiarism detection, or contain hallucinated elements that mimic legitimate research, blurring the line between genuine and fraudulent work. Such practices accelerate error propagation: fabricated citations and ghost references from these sources enter citation networks, appear in downstream legitimate papers via uncritical reuse, and inflate metrics like h-indices while skewing meta-analyses and systematic reviews. Recent large-scale analyses of published literature (2020–2025) have documented repeated invalid citations recurring across independent papers, with an 80.9% surge in invalid citation rates in 2025 alone, highlighting how AI-enabled fabrications create self-reinforcing loops of misinformation that erode trust across disciplines (Xu et al. [16]). In restricted-access environments, where verification tools and global databases are limited, these cascades compound exponentially, forcing researchers to waste resources debunking propagated errors and further disadvantaging international

competitiveness.

8. Practical mitigations and challenges in restricted regions

Practical mitigations for users remain essential: always cross-verify AI-suggested references directly in Scopus, Web of Science, PubMed, or publisher portals; treat LLMs strictly as idea generators or drafting aids, never authoritative sources; and prioritize manual literature searches for any work intended for publication. For energy researchers in restricted regions, these challenges are particularly acute: a hallucinated citation about solar panel efficiency or wind turbine performance that goes uncorrected could lead to flawed local energy policy or misallocated infrastructure investment. When mitigation is unworkable, for example in jurisdictions with heavy internet restrictions—where VPNs are rapidly blocked, international scholarly communication is surveilled or prohibited, and domestic alternatives offer only fragmented or biased coverage of global research—these challenges become near-insurmountable for researchers aiming to produce competitive, internationally oriented scholarship. Whether the result of resource constraints or deliberate design, jurisdiction-dependent hallucinations present a particularly acute challenge for researchers in restricted regions attempting to produce internationally competitive work, where domestic alternatives offer only fragmented coverage. The consequence is a two-tier scholarly ecosystem: well-resourced regions benefit from more reliable AI outputs, while others face mounting—and sometimes insurmountable—barriers to verification. Uneven access to advanced AI infrastructure and tools across different countries can hinder the ability of researchers in less-connected regions to produce and verify scholarship at the same level as their counterparts in AI-rich nations. This divide is further compounded by differential performances across paid versus free AI systems: while premium tiers (e.g., ChatGPT Plus/Pro, Claude Pro, Gemini Advanced) often provide access to more advanced models with improved grounding, refusal mechanisms, or lower hallucination rates in structured tasks (sometimes as low as ~3% for leading paid variants like certain Claude models), independent benchmarks in 2025–2026 have shown that paid versions can occasionally underperform free counterparts in specific open-domain or search-augmented scenarios, with some paid search modes exhibiting higher error rates (e.g., 45–94% in certain evaluations) due to overconfidence or architectural tradeoffs—underscoring that subscription access alone does not guarantee superior factual reliability without rigorous user verification (Vectara [12] [13]; Ross [17]; Artificial Analysis [18]; Brinsa [19]).

9. Broader societal implications and historical parallels

We invoke Orwell's "1984" not to claim equivalence between government censorship and academic publishing practices, but as a cautionary analogy highlighting how concentrated control over information flows—regardless of the specific institutional form—can undermine epistemic integrity. This section should be read as a warning about potential trajectories, not as a description of currently-occurring policy. No evidence is presented that any publisher systematically engages in the kind of 'doublethink' described by Orwell. While this analysis refrains from alleging deliberate AI manipulation, accepting instead that severe hallucinations can arise from restricted information access—the possibility that some jurisdictions intentionally design AI systems to produce falsehoods on certain topics is certainly not ruled out. Schick [20] warns that persuasive AI hallucinations threaten democratic institutions by spreading convincing false narratives, eroding public trust as the line between fact and synthetic fiction blurs. Cazzamatta and Sarisakaloglu [21] document the growing challenge of AI-generated misinformation through cross-national analysis, highlighting how generative AI enables content manipulation at scale while fact-checking methods lag behind. The general use of AI to manufacture consent is certainly something we should think about, but I prefer not to further elaborate on this in this perspective, defocusing from the goal to

limit the spread of falsehood in scientific publishing by AI technicalities. The use of AI to shape public perception is a critical issue for societal governance, but a detailed exploration lies beyond the scope of this paper, which is focused on technical solutions for scholarly publishing. In Orwell's dystopian masterpiece *"1984"* (Orwell [22]), the Party's Ministry of Truth embodies the ultimate authoritarian nightmare: a regime that rewrites history daily, enforces Newspeak to shrink the boundaries of thought, and floods the population with state-sponsored "fake news" until objective reality itself dissolves and the citizenry learns to love Big Brother. This serves as an extreme analogy, not a literal comparison. Echoing this warning with clinical precision, journalist Matt Taibbi has documented the rise of the "censorship industrial complex"—a sprawling public-private hydra of government agencies, Big Tech platforms, NGOs, and academic "disinformation" labs that, as revealed in the Twitter Files and multiple congressional hearings, colludes to pre-bunk, deplatform, and algorithmically bury inconvenient facts, effectively installing a Ministry of Truth for the digital age (Schmidt et al. [23]). Academic analysis of these trends further explores the implications of concentrated control over information flows in the digital public sphere. Similar works, from Herman and Chomsky's *Manufacturing Consent* (Herman and Chomsky [24]), which laid bare the propaganda filters that manufacture "necessary illusions," to Zuboff's *The Age of Surveillance Capitalism* (Zuboff [25]) and the broader literature on narrative warfare, map the same trajectory: a world sliding into soft totalitarianism where curated falsehoods, not ballots or bayonets, become the primary instrument of rule. This accelerating societal drift toward a fake-news-ruled authoritarianism is, however, another story from the issue highlighted here, which must have a quick fix: robust, low-friction technical and procedural safeguards.

10. Publisher-led solutions and current implementations

Moving to the PROCESSING stage of Fig. 1, the most effective long-term solution lies with publisher-led verification systems. The most effective long-term solution to AI-induced falsehoods in scientific publishing lies with AI and the publishers themselves, who profit directly from the volume and speed of publications. Responsibility should not rest solely on individual authors. Mandatory watermarking of AI-generated submissions, automated statistical integrity checks, and rapid author-verification protocols must be rolled out immediately to protect the peer-review system against the growing flood of synthetic papers and fabricated data—content that has proven fiendishly difficult to detect and block effectively at scale. Academic publishers, long criticized for prioritizing profits over openness and equitable access, are already deploying AI more systematically and pervasively than most individual authors across the scholarly communication pipeline—from automated manuscript screening and intelligent reviewer matching to advanced plagiarism detection, metadata enrichment, enhanced content discoverability, and citation analysis (PublishingState.com [26]). While these tools deliver clear benefits in speed, consistency, and cost reduction, they also concentrate significant power in the hands of a few dominant entities, who now control not merely access to published knowledge but the underlying algorithms that process, filter, prioritize, and rank scholarly content (PublishingState.com [26]). Although these AI integrations carry their own risks, further refinement and targeted expansion of such existing publisher-led tools—particularly for integrity checks, hallucination detection, and verification—offer the most immediate and practical path to stemming the growing influx of falsehoods, fabricated citations, and synthetic content threatening the integrity of peer review for papers and grants. Publisher-led verification is not a new concept; it builds on a foundation of pre-existing tools. Elsevier, for example, integrates an automated reference-checking service, MaRS (Machine Learning Reference Service), into its Editorial Manager submission platform for many journals. This is a rule-based system, not a generative AI, but it serves as a crucial precursor. Upon upload of a properly formatted Word manuscript, the system extracts

references and validates journal citations against Scopus and CrossRef, providing clickable links for editors. Validated citations receive direct database links; unvalidated ones are flagged. While this represents a valuable step toward automated integrity checks, the service is imperfect—legitimate Elsevier journal articles are sometimes incorrectly flagged, and database errors naturally propagate. Springer Nature launched a dedicated AI tool in April 2025 to detect irrelevant or problematic references across nearly all its journals and books, complementing existing integrity systems like Geppetto for AI-generated text and SnappShot for image manipulation (Springer Nature [27]). Wiley and Elsevier have similarly expanded AI-assisted editorial checks and mandatory disclosure policies in 2025, requiring authors to declare generative AI use while prohibiting AI as co-authors. These tools represent a shift toward proactive, submission-stage intervention—exactly what is needed at scale. Yet current implementations remain imperfect and inconsistently applied; many journals still rely on post-publication discovery of errors. Expanded, standardized deployment with real-time CrossRef/DOI validation, uncertainty scoring, and hybrid human–AI review could reduce fabrication rates dramatically while preserving publication speed.

11. Limitations of current hyperlinking practices in references

At publication, major publishers (e.g., Elsevier, Springer Nature, Wiley) typically embed hyperlinks in every reference-list entry, usually via DOIs or direct links to databases such as Scopus, CrossRef, PubMed, or Google Scholar. These links aim to improve discoverability and one-click access to cited sources. However, the system is imperfect. Broken links (“*link rot*”) affect a significant portion of references. Recent estimates show up to 37% of DOIs in sampled scholarly content fail to resolve, with higher rates in humanities/business fields and for older citations (Sadatmoosavi et al. [28]). Other studies report lower DOI loss rates around 1.7%, while general URLs experience 5.9% averaged loss, increasing with article age (Briney [29]). In medical journals, the proportion of inaccessible web resources rose from 27.2% to 41.9% three to five years post-publication (Lemaire et al. [30]). Misattributed links occur when automated matching connects a citation to the wrong paper due to similar titles, metadata errors, typos in submitted references, or algorithmic mistakes (especially common with Google Scholar links, where false positives reach 29.9% and author errors affect 19.4% of citations) (Sauvayre [31]). Such erroneous attributions in Google Scholar profiles, often due to common surnames, can skew h-index leaderboards by crediting unrelated works. Incomplete linking leaves many valid references without hyperlinks or flags them incorrectly as unvalidated, even for legitimate journal articles. These issues arise from reliance on author-supplied metadata, variable database coverage, and the dynamic nature of online resources.

12. Proactive mitigation as the path forward

The rapid adoption of generative AI in scientific publishing is introducing increasing amounts of falsehoods into the scholarly record—fake references being just one symptom, alongside fabricated data, invented results, and mass-produced low-quality papers. Banning AI is however neither practical nor desirable, as these tools are becoming essential for research tasks and will only grow more integrated. The effective solution is proactive mitigation: deploy specialized AI verification tools at submission to automatically flag fake or hallucinated references. Existing efforts—such as Elsevier’s reference checker and Springer Nature’s AI-assisted integrity screening—should be expanded, refined, and standardized across all major publishers. These should combine real-time database queries, RAG grounding, and uncertainty detection, with human editorial review for flagged cases. Authors should not be penalized for responsible AI assistance (with disclosure), but publishers—who profit from publication volume—have no less responsibility to protect the literature they disseminate. Only through

robust technology, editorial oversight, and clear policies can we preserve the integrity of scientific publishing in the generative AI era and prevent cascading errors from undermining trust in science. Looking forward, the next 2–3 years will be decisive. Without mandatory publisher-led verification layers at the time of submission, citation cascades from 2025 to 2026 could distort entire subfields, particularly in fast-moving areas like AI itself. Conversely, coordinated investment—perhaps through cross-publisher consortia sharing validation models—could reduce hallucinated citations entering the record by 80–90%, based on early RAG benchmarks. Funding bodies should tie grants to use of certified verification tools, and journals should publish annual “*integrity audits*” disclosing hallucination detection rates. Ultimately, preserving scholarly trust requires moving beyond reactive fixes to a proactive ecosystem where technology serves verification, not just generation. The alternative is a slow erosion of science’s self-correcting foundation at precisely the moment we need it most (Samanta & Chakraborty [3]; Shmatko et al. [9]). While publishers bear primary responsibility for deploying robust AI verification tools to prevent the spread of falsehoods, the scientific community must remain vigilant in overseeing how these tools are implemented and governed. The same AI systems designed to detect hallucinations and fabricated references can, if left unchecked, be repurposed to serve corporate or political interests—prioritizing profit, suppressing inconvenient findings, amplifying preferred narratives, or marginalizing independent scientists whose work challenges established orthodoxies or powerful elites. Without transparent oversight, open auditing mechanisms, and meaningful input from the research community, publisher-led verification risks becoming yet another instrument of control rather than a safeguard for integrity. It is therefore incumbent upon scholars, learned societies, and research institutions to hold publishers accountable: demanding algorithmic transparency, independent validation of detection systems, clear appeals processes for flagged submissions, and protection against discriminatory or politically motivated filtering. The goal is not merely to cleanse the literature of AI-generated fabrications, but to ensure that the tools deployed to protect scholarly integrity do not themselves become vehicles for censorship, commercial exploitation, or the entrenchment of intellectual monopolies.

13. AI as guardian and instrument of control: A cautionary example

Alongside the technological investments, a troubling countercurrent has emerged: the use of AI-assisted systems as potential instruments of control. This section presents a cautionary example, not as established fact about systemic misconduct, but as an illustration of a pathway that requires vigilance. We emphasize that traditional desk-rejections (predating AI) are not the concern; rather, the emergent practice of algorithmic, non-appealable, no-human-signature rejections represents a qualitative shift. For example, there have been documented instances where a publisher issued rejections with a formulaic statement paraphrased as follows:

“After careful evaluation by the publisher, [publisher name here omitted], we regret to inform you that your manuscript does not meet the editorial standards for publication in this journal. The decision is final, with no opportunity for resubmission or appeal.”

The correspondence bore only a corporate signature, the [publisher name here omitted], devoid of any human name—academic or otherwise. This is not a claim that such practices are universal or systematically intended as gatekeeping; rather, it illustrates how AI-assisted screening systems could be deployed to reject manuscripts without peer review, bypassing scholarly judgment. The form letter is a symptom; the instrument of control would be the underlying algorithmic screening system. While there is no proof that this type of corporate censorship is enforced through AI or some layer of unnamed bureaucrats, this is something academics should take into consideration seriously.

Several important qualifications apply. First, traditional form-letter

rejections have existed long before AI chatbots and are not inherently problematic when issued by named editors with clear rationales. Second, the available evidence suggests this practice is not universal across all journals from any given publisher but has been documented for a subset. Third, we present this as a cautionary observation requiring further empirical investigation, not as a claim of systematic misconduct. Fourth, this example is cited for scholarly discussion purposes and should not be construed as an allegation of intentional gatekeeping against any specific publisher.

The receipt of such a rejection letter, particularly one that declares the decision “*final, with no opportunity for resubmission or appeal*,” raises concerns because it potentially contradicts stated publisher policies on appeals.

Editorial policies by [publisher name here omitted] state that appeals against rejection “*should in first instance be handled by the Editor-in-Chief (s) responsible for the journal and/or the Editor who handled the paper*” [32]. This policy further specifies that when an author appeals, “*The Editor-in-Chief or Handling Editor considers the authors' argument, the reviewer reports and decides whether the decision to reject should stand*” [33]. A letter with a corporate signature that pre-emptively declares a decision “*final, with no opportunity for resubmission or appeal*,” such as the formulaic statement proposed earlier, effectively nullifies this stated appeals process by denying authors any meaningful opportunity to engage with the responsible editor.

The Committee on Publication Ethics (COPE), which many publishers officially follow [34], emphasizes that editorial decisions should be made by identifiable editors who take responsibility. When a rejection bears only a corporate signature, authors cannot determine who made the decision or address concerns to the appropriate person. This section serves as a cautionary illustration of potential risks rather than an assertion that any specific publisher systematically employs AI to engage in such practices. It is already a fact that publishers reject manuscripts in their own name, declaring the decision final with no opportunity for resubmission or appeal; the only uncertainty lies in what role AI may have played in that decision. Although further empirical research is necessary to determine the prevalence and intent behind these practices, there is no doubt that the academic community must remain extremely vigilant.

14. Specifying priorities for encoding in AI systems

The statement that “*AI systems encode the priorities of those who build them*” requires concretization. Below, we specify what those priorities should be across four scholarly stages, with particular attention to the energy sector.

Table 1 summarizes the priorities that should be encoded into AI systems for scholarly publishing. For literature searches, the priority is accuracy over recall, implemented by validating against Scopus and CrossRef and flagging uncorroborated citations. For paper writing, the priority is transparency, implemented through mandatory AI disclosure and real-time uncertainty estimation. For manuscript reviewing, the priority is appealability, implemented through named editors, clear rationales, and the rejection of “*final, no appeal*” decisions. For journal editing, the priority is equitable access, implemented through free

verification tools and the avoidance of institutional bias in screening.

In energy research, sector-specific priorities emerge. AI-generated citations about battery degradation rates, cycle life, or energy density must be validated against peer-reviewed sources such as the Journal of Power Sources or Nature Energy before inclusion in literature reviews. Efficiency claims for novel solar photovoltaic materials like perovskites require verification against NREL or equivalent databases, as AI hallucinations about “*record efficiency*” are a documented risk. For carbon capture, cost estimates in dollars per ton of CO₂ and capture rates require grounding in pilot plant data, not AI-generated extrapolations. Citations about grid reliability, including blackout probabilities or renewable integration thresholds, must be traceable to utility reports or peer-reviewed models.

Table 2 summarizes how these priorities differ for energy versus other sectors. In energy, the key risk involves infrastructure investment and climate policy, so the mitigation priority is domain-specific validation layers for cost and efficiency figures. In medicine, the key risk involves patient safety and treatment efficacy, requiring regulatory-grade verification aligned with FDA or EMA standards. In law, the key risk involves legal precedent and court filings, demanding zero tolerance for fabricated citations due to sanction risks. In AI and computer science, the key risk involves self-referential citation cascades, requiring cross-disciplinary validation to prevent models from citing themselves.

Implementing these priorities would mitigate the issues discussed throughout this article in four ways. First, it would reduce hallucination propagation during the WRITING stage via real-time database validation. Second, it would catch errors before publication during the REVIEWING stage via submission-stage screening. Third, it would prevent algorithmic gatekeeping during the PROCESSING stage via named editor review of AI flags. Fourth, it would ensure equitable access across regions by providing free verification tools to researchers in restricted jurisdictions.

15. Conclusion: The inevitable trajectory for AI

Whether embraced or resisted, artificial intelligence is irrevocably reshaping scientific publishing—for better and for worse. On one hand, AI offers unprecedented capabilities: accelerating literature synthesis, democratizing access to knowledge, automating editorial workflows, and enhancing discoverability. On the other, as this paper has documented, it introduces systemic vulnerabilities: hallucinated citations, fabricated datasets, and the proliferation of paper-mill manuscripts that threaten the scholarly record. But beneath this technical duality lies a deeper, more troubling dimension: the risk that AI in publishing could become a mechanism for elite control, ideological gatekeeping, and the systematic silencing of dissent. We emphasize that this is a risk and a cautionary observation, not an established fact. However, the documented emergence of non-appealable, publisher-only rejections (as discussed in Section 13) illustrates a pathway by which such practices could develop. There is currently no systematic evidence that major publishers intentionally deploy AI-assisted screening as a tool of gatekeeping rather than quality control; we present this as a theoretical concern requiring empirical investigation. When major publishers issue categorical, non-appealable rejections signed only by the corporate

Table 1
Priorities that should be encoded into AI systems for scholarly publishing.

Stage	Priority	Implementation
Literature searches	Accuracy over recall	Validate against Scopus/CrossRef; flag uncorroborated citations
Paper writing	Transparency	Mandatory AI disclosure; real-time uncertainty estimation
Manuscript reviewing	Appealability	Named editors; clear rationales; no “final, no appeal” decisions
Journal editing	Equitable access	Free verification tools; no institutional bias in screening

Table 2
How these priorities differ for energy versus other sectors.

Sector	Key Risk	Mitigation Priority
Energy	Infrastructure investment, climate policy	Domain-specific validation layers for cost/efficiency figures
Medicine	Patient safety, treatment efficacy	Regulatory-grade verification (FDA/EMA alignment)
Law	Legal precedent, court filings	Zero-tolerance for fabricated citations (sanction risks)
AI/CS	Self-referential citation cascades	Cross-disciplinary validation (models citing themselves)

entity—bypassing peer review and named editors—there is a risk that they deploy AI-assisted screening as a tool of gatekeeping, not quality control. If a rejection declares itself “*final, with no opportunity for resubmission or appeal*” and carries no human signature, this would represent administrative suppression dressed in algorithmic legitimacy, rather than genuine editorial judgment. This matters because AI systems encode the priorities of those who build them. As we detail in Section 14 below, these priorities require explicit articulation. When deployed in editorial workflows, they can be calibrated—explicitly or implicitly—to favor established networks, high-status institutions, commercially aligned perspectives, and methodologies that pose no challenge to the status quo. Conversely, they could flag as “*problematic*” research that originates from the Global South, involves “*dissidents*” of narratives promoted by the Global North, employs critical frameworks, or questions prevailing orthodoxies. Unlike overt censorship, algorithmic gatekeeping operates in the shadows. Authors may receive form letters, not rationales. They may be unable to appeal to a committee of peers. Over time, this chilling effect could discipline entire fields, steering scholarship away from uncomfortable questions and toward safer, more palatable terrain. None of this negates the need for AI verification tools to detect hallucinations and fabricated citations. On the contrary, robust verification is essential. But integrity is not merely about factual accuracy—it is also about openness, contestation, and the freedom to pursue knowledge without fear of extra-scientific sanction. When tools designed to catch errors are repurposed to enforce conformity, the cure becomes worse than the disease. What is needed, therefore, is not the abandonment of AI but its embedding within a governance framework that prioritizes scholarly autonomy. Publishers must disclose screening criteria, ensure AI-flagged rejections are reviewed by named academic editors, and provide meaningful appeals processes. Independent auditing of editorial AI systems—by learned societies or research funders—should be mandatory. And the scholarly community must reclaim stewardship through support for non-profit, community-led publishing models. Like it or not, AI is now woven into the fabric of scholarly publishing. The same technologies that could safeguard the literature against hallucinations could also be used to insulate established power from critical scrutiny. Whether they become tools of integrity or instruments of control depends entirely on whether the scientific community acts now—with clarity and collective resolve—to demand transparency, accountability, and the preservation of academic freedom in the age of algorithmic gatekeeping. Publishers may hold the technical keys, but the scientific community must retain ultimate stewardship of the scholarly record. If we entrust the guardianship of scholarly integrity solely to commercial publishers—however sophisticated their AI tools—we risk not only the pollution of the literature with fabricated citations but also the subversion of knowledge itself by the very institutions that claim to preserve it. The defense of science, then, begins not with better algorithms alone, but with the collective refusal to outsource intellectual freedom to corporate interests.

16. Commentary on the nature news feature (1 April 2026)

This manuscript and the Nature News Feature [35] speak directly to each other in fascinating ways—sometimes aligning, sometimes revealing tensions which are worth highlighting.

The Nature analysis grounds several claims from this manuscript in data. Examining ~4000 publications from five major publishers, 65 of the 100 most suspicious papers contained at least one invalid reference. Extrapolated to ~7 million 2025 publications, this suggests over 110,000 may contain invalid references—quantitative support for what this manuscript calls “error cascades” and “erosion of trust.”

There are key echoes in the two works. GPT-4o produced ~20% fabricated references (this manuscript notes GPT-4 reduced to 18% from GPT-3.5's 55%); “Frankenstein citations” match this manuscript “compound deceptions”; screening pilots show publishers catch an “order of magnitude” more flags than published analyses; and 2.6% of 2025 CS

conference papers had potential hallucinations (up from 0.3% in 2024). This manuscript observation that tools are “imperfect and inconsistently applied” is supported by Nature's acknowledgment of false positives and that flagged references aren't always genuinely problematic.

There are also key tensions in between the two works: This manuscript warning that AI verification risks becoming “instruments of control” is absent from the Nature piece, which presents publisher-led screening as unalloyed good. Using tools like Elsevier MaRS, Springer-Nature could easily check references if they really wanted to solve the issue. Otherwise, it looks like a pretext for opaque control (censorship/misinformation) under the guise of fighting misinformation. Existing tools are already sufficient to address most of the issues; failure to deploy reflects lack of will, not capability. The professed motivation may be pretext; selective deployment enables discretionary gatekeeping. This manuscript praises existing tools as valuable but documents serious limitations—DOI link rot up to 37%, misattributed links (29.9% false positives for Google Scholar), incomplete indexing, database errors. The Nature commentary confirms tools fail due to formatting variations and missing DOIs. “Easily check” overstates reality. More easily than manual? Yes. Easily? No—not with acceptable false-positive rates across heterogeneous formats.

This manuscript raises concerns about “algorithmic, non-appealable, no-human-signature rejections.” Section 15 argues AI systems encode builders' priorities and tools could be repurposed for conformity. The entire framing of AI misinformation as a problem may be itself a control mechanism. While this manuscript is careful: “this is a risk and a cautionary observation, not an established fact” with “no evidence that any publisher systematically engages” in such practices, it warns the scientific community to be vigilant.

There is a possible dichotomy—publishers either solve detection or use it as pretext. This manuscript supports a third more likely possibility: imperfect tools, technical limitations, and economic incentives (speed, volume) that conflict with rigorous verification. “Easily check” is unsupported—35 of 100 most suspicious papers were false positives; at scale, even 10% false positives create massive friction. Equating verification with censorship conflates detecting fabrications (serves integrity) with non-appealable algorithm-driven rejections (threatens integrity). This manuscript distinguishes these.

Skepticism about the selective deployment deserves attention. If Springer Nature has an AI tool (launched April 2025) and IOP Publishing now uses Veracity across all journals, why aren't these tools mandatory and universal? And why Elsevier MaRS is not consistently used across all the Elsevier Journals? Possible explanations include technical integration costs, concern about false positives slowing publication (economic pressure), lack of standardized benchmarks, or selective deployment enabling discretionary filtering. This manuscript Section 13 example (final, non-appealable rejection with no human signature) would support the suspicion if AI hallucination detection was shown to be the stated justification.

Institutions profiting from publication volume are now asked to deploy verification tools that could slow publication and give them discretionary power. Skepticism about this coincidence of interest is not irrational. But the leap from “this could be abused” to “this is necessarily a pretext for censorship” is still evidentially unsupported. More importantly, it forecloses the very governance solution this manuscript proposes:

- transparent oversight,
- named editors,
- appeals processes,
- independent auditing.

This manuscript holds both truths together—verification is necessary AND verification systems require democratic governance.

CRediT authorship contribution statement

Alberto Boretti: Writing – review & editing, Writing – original draft, Data curation, Conceptualization.

Declaration of competing interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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